DSE4101 Capstone Project in Data Science and Economics I

College of Humanities and Sciences
National University of Singapore

Semester 1, AY 2026/2027

1 Coordinator and Class Meeting Schedules

Coordinator	Ta-Cheng HUANG (ECON)
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Office	AS2 05-07
Office Hours	By appointment

The class will meet in person on the following dates at **AS2-0312**.

- Tuesday, 11 Aug 2026, 6 pm – 7 pm
- Tuesday, 15 Sept 2026, 6 pm – 10 pm
- Tuesday, 3 Nov 2026, 6 pm – 10 pm

2 Course Description and Objectives

2.1 Course Description

This course allows students to apply theories, concepts and methodologies taught in various integrated data science and economics courses to an pre-approved self-directed project. Students may, for example, work on projects that deepen their understanding of certain aspects of digital currencies, FinTech or the digital economy, or that involve data-driven problem-solving or a detailed case study within a practical setting in economics, finance, public policy or society. Each project culminates in a written report detailing motivation, approach, methods and findings.

2.2 Course Objectives

Upon completion this course, students will be able to:

- Showcase the knowledge and analytical skills that they have acquired in other data science and economics courses.
- Demonstrate an awareness of data-scientific problems of practical interest in economics, finance, public policy and society.
- Develop independent research and problem-solving skills in emerging or rapidly developing technologies or applications.

2.3 Prerequisite

- DSE3101 Practical Data Science for Economics AND
- EC3304 Econometrics II

3 Course Structure and Policies

3.1 Course Structure

This course provides students with the opportunity to undertake a self-directed, instructor-approved project in one of three domains: FinTech, digital currencies, or the digital economy.

From Weeks 1 to 6, students will first self-select a domain of interest. Within each domain, students will be randomly assigned to teams. Each team will then select a group project topic from a list provided by the coordinator. To encourage early planning and steady progress, each group must submit a short written proposal by Tuesday of Week 3. This Phase 1 Proposal should explain the project's motivation, identify the key literature, provide a basic work plan, and outline each member's responsibilities.

Before the Week 6 presentations, each student will be assigned another group's presentation to review. To help the assigned discussants prepare thoughtful questions and feedback, each group must submit its presentation slides by Sunday of Week 6. On Tuesday of Week 6, groups will present their initial findings and conceptual framework, and each discussant must submit a written review report by the following Tuesday.

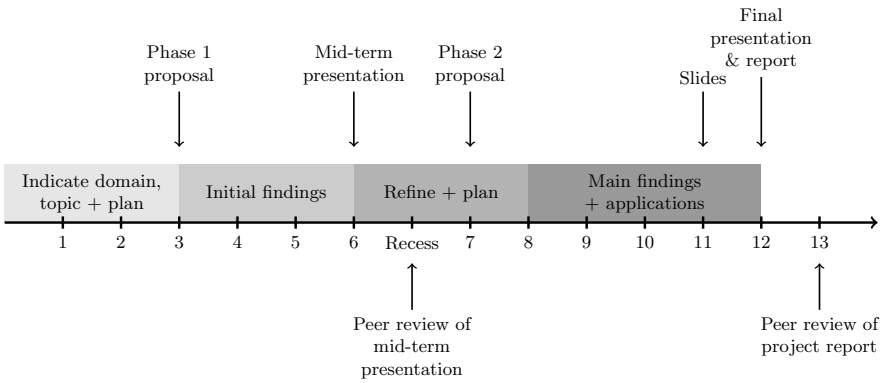
From Weeks 7 to 12, teams will apply their proposed methodologies to real-world data. To ensure the research remains on track, each group must submit a Phase 2 Proposal by Tuesday of Week 7. This updated proposal should refine the research question, if necessary, finalize the data sources and methodology, and provide a more detailed plan for completing the project.

To help the assigned discussants prepare thoughtful questions and feedback, each group must submit its presentation slides by Sunday of Week 12. On Tuesday of Week 12, groups will deliver their final presentations. Each group must then submit its written report by Friday of Week 12. Each discussant must submit a final review report on the assigned group's written report by Friday of Week 13.

The table and figure below summarize the key events and assignments.

Week	**Date**	**Time**	**In-Class Event**[a]	**Task**[b]
1	11 Aug	6–7 pm	Course briefing	
1	14 Aug			Indicate domain preferences
2	18 Aug			Topic selection
3	25 Aug			Phase 1 proposal due
6	13 Sept			Mid-term presentation slides
6	15 Sept	6–10 pm	Mid-term presentation	
R	22 Sept			Review report on a presentation
7	29 Sept			Phase 2 proposal
12	1 Nov			Final presentation slides
12	3 Nov	6–10 pm	Final presentation	
12	6 Nov			Final group project report
13	13 Nov			Review report on a group project

[a]Meeting in person at AS2-0312.
[b]Tasks are due by 11:59 pm on the deadline, unless stated otherwise.



3.2 Grades

Student performance will be assessed based on group and individual components, with an emphasis on collaborative output and individual engagement. The grading breakdown is as follows:

Components	**Weightage**
Group (60%)	
Phase 1 Proposal	5%
Mid-term Presentation	10%
Phase 2 Proposal	5%
Final Presentation	15%
Group Report	25%
Individual (40%)	
Participation: Discussants	10%
Review Report: Peer Mid-term Presentation	10%
Review Report: Peer Group Project	20%
Total	100%

Proposals. Two (2) group project proposals together account for 10% of the course grade, with 5% allocated to each.

1. The **Phase 1 Proposal**, due in Week 3, accounts for 5% of the course grade. This short written document should clearly articulate the project's motivation, list key literature, outline each member's responsibilities, and provide a basic work plan. It serves to ensure the group starts the project with a clear direction and realistic goals. This sets the foundation for the mid-term presentation.
2. The **Phase 2 Proposal**, due in Week 7, accounts for 5%. This updated proposal should justify the chosen dataset, define the research questions and methods, describe expected outcomes, assign tasks, and lay out a timeline for completion.

Presentations. Two (2) group project presentations constitute 25% of the final grade, with 10% allocated to Mid-Term Group Presentation and 15% the Final Group Presentation.

1. The **Mid-Term Group Presentation**, delivered in Week 6, contributes 10% to the final grade. Each group will present their understanding of the relevant literature and the

conceptual framework underpinning their project. The clarity, coherence, and relevance of the presentation will be key grading criteria.

2. The **Final Group Presentation**, held in Week 12, is worth 15% of the grade. The presentation should demonstrate the application of theoretical and empirical tools to real-world data, provide insightful interpretation of findings, and communicate results effectively.

Participation as Discussants. Serving as a discussant is an integral part of this course and contributes 10% to the final grade. As discussants, students are expected to engage actively during the Mid-term and Final Group Presentation sessions by asking thoughtful questions, offering constructive feedback, and responding professionally during peer discussions.

Final Report. The Final Group Report, due after the Week 12 presentation, makes up 25% of the course grade. This report should provide a detailed and rigorous discussion of the project, including motivation, literature, data description, empirical strategy, findings, and conclusions.

Review Reports. Each student is required to submit two review reports — one following the mid-term group presentation and another after the final group project. These reviews together account for 30% of the course grade.

1. The **Mid-Term Review Report on a Peer Group Presentation**, due in Week 7, is worth 10%. Each student will be assigned to a presentation and serve as a discussant. The report should succinctly summarize the main findings of the assigned presentation and provide constructive feedback on the clarity, coherence, and relevance of the content.
2. The **Final Review Report on a Peer Group Project**, due in Week 13, contributes 20% to the final grade. In this assignment, each student will be assigned a group project report from another domain. The report should summarize the key ideas and findings of the project and offer a thoughtful critique of its research design, data usage, methodology, and conclusions. Students are expected to assess both the technical execution and the clarity of communication.

Peer Evaluation of Group Contributions. While the group components are graded based on the quality of the group's work, each student's final score for the group components may be adjusted to reflect their individual contribution. At the end of the semester, all students will complete a confidential peer evaluation of their teammates' contributions. If at least half of a student's teammates report that the student did not contribute adequately to the group project, the student's score for the group assessment components will be adjusted. To ensure fairness, any claim of inadequate contribution must be supported by evidence, such as communication records, version history, meeting attendance, task allocation records, or other relevant documentation.

Your grade will be based on your performance over the semester. The previous six paragraphs (i.e., proposals, presentations, participation, final report, review reports, and peer evaluation) set out the *only* method that will be used to determine course grades. If you feel that personal circumstances are affecting your academic performance, or are concerned that your work is not going to earn you the grade that you require for some purpose (e.g., getting/keeping an assistantship or a scholarship), please explore your options and take appropriate action in good time (see Important Dates below).

3.3 Canvas

All course materials and announcements will be posted on Canvas (https://canvas.nus.edu.sg). You should check Canvas on a frequent and consistent basis.

3.4 E-mail

It is every student's responsibility to check their NUS official e-mail on a frequent and consistent basis. Announcements will be sent to your NUS e-mail. Any e-mail to the coordinator should be initiated from NUS e-mail with the course number, i.e., **DSE4101**, in the subject line.

University Policies

Important Dates When a student withdraws from a module between the first day of Week 3 of the instructional period and the last day of the Recess Week, a 'W' grade will be recorded in the transcript. Withdrawals after this period (i.e., from the first day of Week 7 of the instructional period) will result in an 'F' grade, which will be included in the computation of the CAP and will not be eligible for the S/U option.

Academic Honesty The University is committed to nurturing an environment conducive for the exchange of ideas, advancement of knowledge and intellectual development. Academic honesty and integrity are essential conditions for the pursuit and acquisition of knowledge, and the University expects each student to maintain and uphold the highest standards of integrity and academic honesty at all times.

The University takes a strict view of cheating in any form, deceptive fabrication, plagiarism and violation of intellectual property and copyright laws. Any student who is found to have engaged in such misconduct will be subject to disciplinary action by the University. Such misconduct will include, but is not limited to, the misuse of content or language generated by artificial intelligence (AI) computer programmes.

It is important to note that all students share the responsibility of protecting the academic standards and reputation of the University. This responsibility can extend beyond each student's own conduct, and can include reporting incidents of suspected academic dishonesty through the appropriate channels. Students who have reasonable grounds to suspect academic dishonesty should raise their concerns directly to the relevant Head of Department, Dean of Faculty, Registrar, Vice Provost, or Provost.

Plagiarism Plagiarism is the taking of the work or ideas of another person or entity, including artificial intelligence (AI), and presenting them as one's own without proper attribution. This is a form of academic dishonesty. It undermines fair evaluation of the student's own learning and effort, and it also undermines the system of scholarly research and development of ideas. NUS does not condone plagiarism, regardless of source.

AI Tools ARTIFICIAL INTELLIGENCE TOOLS such as ChatGPT are now essentially costless and do not require special expertise to use. As part of the repertoire of tools available in the modern world, we understand how you–our students–might be drawn to using them. It is also likely employers and stakeholders will soon assume that graduates have a basic competency for using such tools, just like how they now assume that new hires are able to use calculators, spreadsheets, and word processing software. That said, there are some things we need you to take to heart.

FIRST, while we are fine with you experimenting with new tools, we are also wary you will end up taking shortcuts that will disadvantage you in the longer run. Consider, for instance, the different levels of capability involved in the following three scenarios:

1. Using a tool to generate an output wholesale from inputs.
2. Using a tool to generate intermediate outputs that are then developed into a final output through further human intervention without the use of the tool.
3. Evaluating the output of a tool to confirm its accuracy, relevance, objectivity, and completeness.

You will need a higher level of capability in yourself to be able to do (2) and (3), as compared with doing (1), and conversely, functions that only require (1) are at a higher risk of being completely replaced by AI. The implication is that you are only cheating your future selves if you go straight to using such tools before learning the actual subject matter. If you, as a learner, take shortcuts today, you risk becoming first in line to be replaced by bots!

Since we target learning outcomes that hone higher-level capabilities, and we want our students to demonstrate originality, voice, intellectual engagement with content, rigor, and contribution to discourse, we will often need you to produce unaided work as part of your assessment. This is no different from how, even though we expect you to use calculators and spreadsheets, we still want you to learn the underlying math and computational processes. And even though we are fine with you using search engines, spelling and grammar checkers, or for that matter, lexicons and thesauri, we still want you to be able to independently organize your thoughts and produce your own compositions–so that you can assess the tools' deliverances. SECOND, note that normal academic rules continue to apply. As provided in the Code of Student Conduct:

The University takes a strict view of cheating in any form, deceptive fabrication, plagiarism and violation of intellectual property and copyright laws. Any student who is found to have engaged in such misconduct will be subject to disciplinary action by the University. In this regard, representing an AI's output as your own work is plagiarism. (See "Guidelines on the Use of AI Tools for Academic Work".)

THIRD, do be aware that the tools typically require you to sign up for an account and accept various terms and conditions. We encourage you to be very clear about what you are accepting if you sign up for an account. You should also be aware that there are lawsuits underway challenging the intellectual property provenance of many of these tools. If they are ruled illegal (while this is unlikely, it is not impossible), there may be implications for your use as well. And finally, don't forget that like all online tools, platforms can sometimes go down without warning, or are under such high load that their responses become too slow for use. It is thus unwise to assume that you will always have access.

Additional guidance can be found at:

- NUS Code of Student Conduct
- NUS Academic Integrity Essentials
- NUS Plagiarism Policy
- Guidelines on the Use of AI Tools For Academic Work

Last updated on August 4, 2026